

DSA EXPERIMENT 4:Database Management using arrays of structures

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```
#include <stdio.h>
```

```
struct student  
{  
    int roll;  
    char name[80];  
    float SGPA;  
};
```

```
void create(struct student arr[], int n);  
void display(struct student arr[], int n);  
void search(struct student arr[], int n);
```

```
void create(struct student arr[], int n)  
{  
    printf("Enter student details:\n");
```

```
    for (int i = 0; i < n; i++)  
    {  
        printf("Enter roll no, name and SGPA: ");  
        scanf("%d %s %f",  
            &arr[i].roll,  
            arr[i].name,  
            &arr[i].SGPA);  
    }  
}
```

```
void display(struct student arr[], int n)  
{  
    printf("\nStudent Details:\n");
```

```
    for (int i = 0; i < n; i++)
```

```

    {
        printf("Roll No: %d\n", arr[i].roll);
        printf("Name: %s\n", arr[i].name);
        printf("SGPA: %.2f\n", arr[i].SGPA);
    }
}

```

```

void search(struct student arr[], int n)
{
    int roll, i;

    printf("\nEnter roll number to search: ");
    scanf("%d", &roll);

```

```

    for (i = 0; i < n; i++)
    {
        if (arr[i].roll == roll)
        {
            printf("\nStudent found!\n");
            printf("Roll No: %d\n", arr[i].roll);
            printf("Name: %s\n", arr[i].name);
            printf("SGPA: %.2f\n", arr[i].SGPA);
            return;
        }
    }
}

```

```

    printf("\nStudent not found.\n");
}

```

```

int main()
{
    struct student s[100];
    int n;

    printf("Enter number of students: ");
    scanf("%d", &n);

    create(s, n);
    display(s, n);
}

```

```
search(s, n);

return 0;
}
```

Output

```
Enter number of students: 4
Enter student details:
Enter roll no, name and SGPA: 51 Rushita 9
Enter roll no, name and SGPA: 52 Srishti 9.1
Enter roll no, name and SGPA: 53 Gauravi 9.7
Enter roll no, name and SGPA: 54 Bhakti 9.2
```

Student Details:

```
Roll No: 51
Name: Rushita
SGPA: 9.00
Roll No: 52
Name: Srishti
SGPA: 9.10
Roll No: 53
Name: Gauravi
SGPA: 9.70
Roll No: 54
Name: Bhakti
SGPA: 9.20
```

```
Enter roll number to search: 51
```

Student found!

```
Roll No: 51
Name: Rushita
SGPA: 9.00
```

```
=== Code Execution Successful ===
```